

Evaluation of the AI for Advanced Semiconductor Manufacturing Workshop

Introduction

This report evaluates the *AI for Advanced Semiconductor Manufacturing Workshop* through the lens of adult learning theory and program evaluation. The workshop was designed to provide participants with hands-on experience in semiconductor fabrication, data collection, and artificial intelligence applications within advanced manufacturing contexts. The evaluation focuses on participant needs, engagement, instructional effectiveness, confidence development, and opportunities for future improvement.

The workshop addressed emerging workforce needs in semiconductor manufacturing and industrial AI while supporting learners from diverse educational and technical backgrounds. From an adult education perspective, the program emphasized experiential learning, practical application, and career relevance, all of which are important characteristics of effective adult instruction.

Program Evaluation Purpose

The purpose of this evaluation was to better understand how adult learners experienced the workshop, what instructional approaches supported learning most effectively, and what barriers or challenges participants encountered throughout the learning process.

The evaluation specifically examined:

- participant backgrounds and prior preparation,
- learner confidence and readiness,
- instructional preferences,
- engagement with workshop activities,
- perceived learning outcomes,
- and recommendations for improving future workshop offerings.

Pre- and post-workshop surveys were used to assess learner expectations, confidence, participation, and perceptions of instructional effectiveness.

Adult Learner Analysis

Adult learners typically value instruction that is practical, relevant, collaborative, and directly connected to real-world goals. Unlike traditional classroom learners, adults often enter learning environments with prior experiences, varying confidence levels, and clearly defined educational or career motivations.

The workshop addressed these needs by integrating:

- hands-on semiconductor fabrication,
- AI-based classification tasks,

- real-world manufacturing datasets,
- guided technical instruction,
- and opportunities for independent exploration.

Participants engaged directly with fabrication processes, image datasets, data analysis activities, and machine learning tools. This experiential structure supported authentic learning and encouraged learners to connect technical concepts to practical workforce applications.

The workshop also reflected several principles of Malcolm Knowles' theory of andragogy, particularly:

- self-directed learning,
- experiential engagement,
- problem-centered instruction,
- and immediate practical relevance.

Methodology

The workshop evaluation utilized anonymous pre- and post-surveys to examine participant experiences, instructional effectiveness, learner confidence, and perceived learning outcomes.

Survey questions included:

- Likert-scale items,
- multiple-choice questions,
- and open-ended responses.

These questions were designed to capture both quantitative and qualitative feedback regarding:

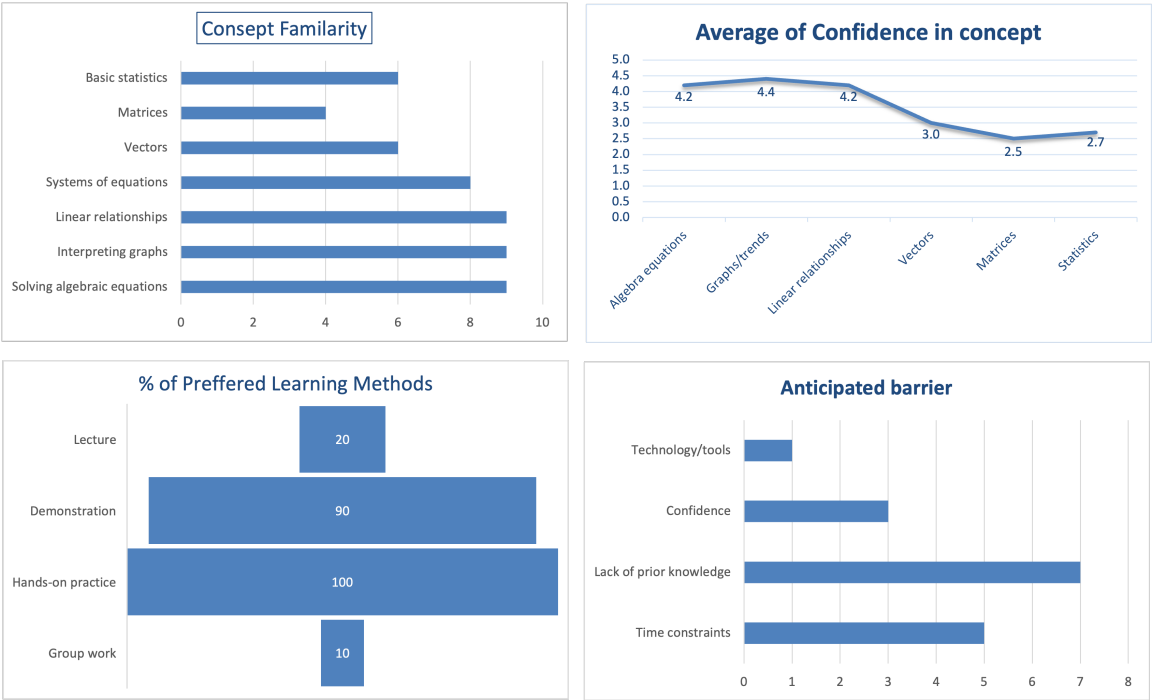
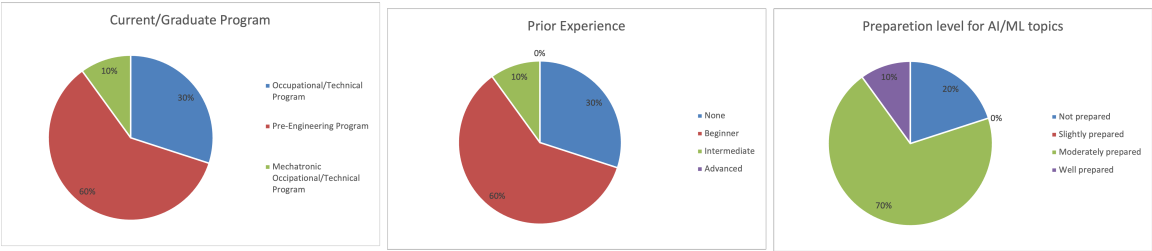
- learner goals,
- instructional experiences,
- barriers to learning,
- confidence development,
- and overall workshop satisfaction.

Quantitative survey responses were analyzed using descriptive statistics and visualized through charts to identify trends in learner preparation, engagement, and perceived outcomes. Qualitative responses were reviewed through thematic analysis to identify recurring themes related to learner motivations, instructional effectiveness, challenges, and recommendations for improvement.

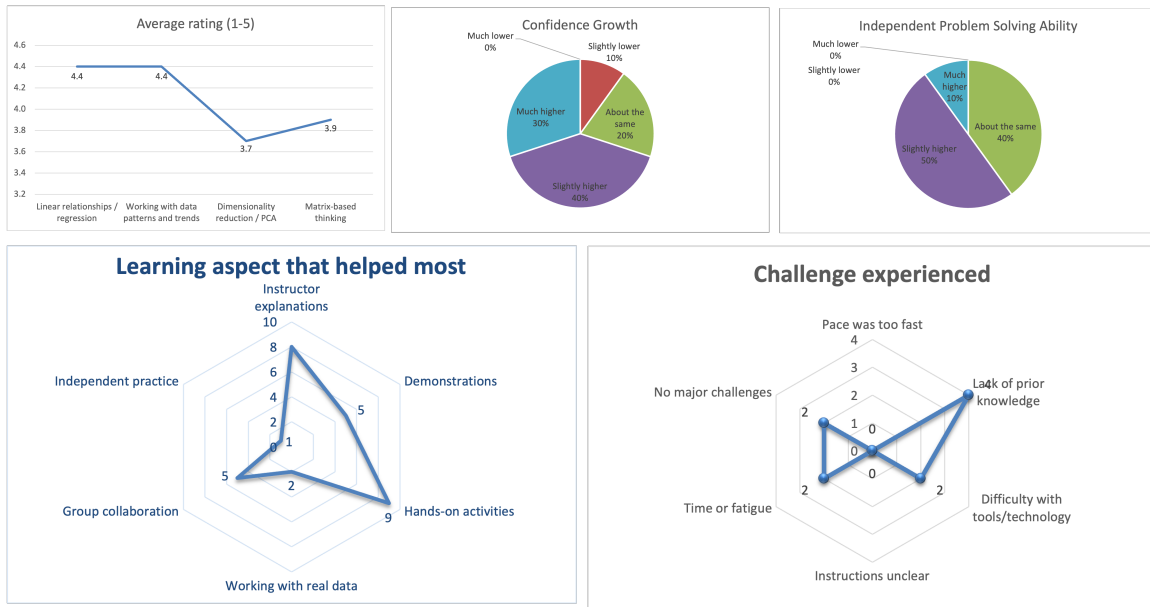
The evaluation focused specifically on understanding how adult learners experienced the workshop and how instructional strategies supported engagement, confidence, and applied learning.

Survey Results and Discussion

Pre Survey



Post Survey



Analysis of Pre-Survey Results

The pre-survey responses revealed that participants entered the workshop with diverse educational and technical backgrounds. Most participants identified as pre-engineering students, while others came from occupational or technical pathways. This diversity demonstrates the importance of designing instruction that is flexible and accessible to learners with varying levels of preparation.

The survey data also showed that participants were generally more familiar with algebraic equations, graphs, and linear relationships than with vectors, matrices, and statistics. These findings suggest that learners possessed foundational mathematical knowledge but had less experience with advanced concepts associated with machine learning and AI applications.

The strong preference for hands-on practice suggests that participants learned most effectively through experiential engagement rather than passive instruction. Demonstrations and instructor-guided activities were also highly valued, while relatively few participants preferred lecture-based instruction alone. These findings reinforce the importance of active learning and contextualized technical education for adult learners.

The most anticipated barrier before the workshop was lack of prior knowledge, followed by time constraints and confidence concerns. This indicates that many learners approached the workshop with uncertainty regarding their ability to succeed in a highly technical environment. These responses highlight the importance of instructional support, scaffolding, and confidence-building opportunities within adult technical education.

The pre-survey open-ended responses revealed that participants were strongly motivated by practical learning opportunities and career-related skill development.

Many learners expressed interest in gaining hands-on experience in semiconductor manufacturing, data analysis, machine learning, and engineering applications.

Participants frequently emphasized the desire to apply workshop knowledge to future careers in:

- robotics,
- automation,
- manufacturing,
- and engineering fields.

Several participants also expressed interest in understanding how AI can be applied to real-world semiconductor manufacturing problems. These responses suggest that learners were highly motivated by the practical relevance of the workshop and its connection to workforce preparation and future educational goals.

Although many participants reported few concerns prior to the workshop, some expressed uncertainty regarding programming experience and technical preparation. These comments suggest that learners entered the workshop with varying levels of confidence and prior knowledge, reinforcing the importance of structured guidance and learner support in technical education settings.

Analysis of Post-Survey Results

The post-survey responses provided insight into participant learning outcomes, instructional experiences, confidence development, and overall perceptions of the workshop.

The findings suggest that the workshop successfully improved participant confidence and understanding of technical concepts. Participants reported stronger understanding of:

- linear relationships,
- data analysis,
- machine learning concepts,
- and applied AI processes.

Confidence levels remained highest in concepts that were reinforced through direct application and hands-on learning activities.

The workshop appeared particularly effective in helping participants build confidence in learning technical material independently. Many respondents indicated that their confidence was either “slightly higher” or “much higher” after completing the workshop. From an adult learner perspective, this outcome is especially important because self-efficacy plays a major role in persistence, motivation, and lifelong learning.

Participants identified hands-on activities, demonstrations, and instructor explanations as the most valuable aspects of the workshop. Activities involving real

data and practical problem solving were also viewed positively. These findings support experiential and learner-centered approaches to technical education.

The post-survey results also demonstrated very high levels of participant satisfaction with the workshop structure and instructional approach. Most participants reported that the balance between guidance and independence was appropriate, with nine out of ten respondents indicating that the workshop provided the right level of instructional support. Only one participant reported needing additional guidance, while no participants indicated a preference for more independence.

These findings suggest that the workshop successfully balanced instructor support with opportunities for self-directed learning, which is a critical component of effective adult education.

Overall workshop evaluations were overwhelmingly positive. Nine participants rated the workshop as “Excellent,” while one participant rated it as “Good.” Similarly, nine participants stated that they would “Definitely” recommend the workshop to others, and one participant responded “Probably.” No participants selected neutral or negative responses regarding recommendation.

These results indicate strong participant satisfaction, perceived value, and relevance of the workshop experience. From an adult learner perspective, these responses suggest that the workshop effectively met learner expectations for practical, meaningful, and career-relevant education.

Although the workshop was evaluated positively overall, participants also identified several challenges. The most common barriers included:

- lack of prior knowledge,
- difficulty with tools or technology,
- and the pace of instruction.

These findings suggest that future workshop iterations may benefit from additional scaffolding, pre-workshop preparation materials, and opportunities for review or reflection during instruction.

The post-survey qualitative responses strongly reinforced the importance of experiential and instructor-supported learning approaches. Participants consistently identified:

- hands-on activities,
- lab work,
- demonstrations,
- and instructor explanations

as the most effective aspects of the workshop.

Several learners emphasized that repeated explanations, practical application, and opportunities to work directly with real data significantly improved their understanding of technical concepts.

Participants also described the instructors as:

- knowledgeable,
- approachable,
- supportive,
- and communicative.

These responses suggest that the workshop fostered a collaborative and psychologically safe learning environment. Adult learners often benefit from supportive instructional spaces where they feel comfortable asking questions, making mistakes, and experimenting with unfamiliar technical skills.

While overall feedback was highly positive, participants also identified opportunities for improvement. Suggestions included:

- providing additional breaks,
- improving computer performance,
- offering more setup guidance for technical tools,
- and supplying preparatory materials for advanced mathematical concepts.

These recommendations indicate that future workshops could further strengthen learner support by incorporating additional technical scaffolding and preparatory resources.

Connection to Adult Learning Theory

The workshop aligns strongly with Malcolm Knowles' principles of andragogy. Adult learners are generally motivated when learning is immediately relevant, problem-centered, and connected to real-world goals. The workshop addressed these needs by engaging learners in authentic semiconductor manufacturing and AI applications rather than abstract theoretical exercises.

The emphasis on small-group instruction and guided practice also supported adult learner needs by creating a collaborative and supportive environment. Adult learners often bring prior experiences, anxieties, and varied confidence levels into technical learning spaces. The workshop's balance between instructor guidance and independent exploration helped participants gradually build competence while maintaining engagement.

Experiential learning was one of the strongest features of the program. Rather than relying primarily on lecture-based instruction, participants actively worked with:

- fabrication processes,
- image datasets,
- AI tools,
- and machine learning workflows.

This instructional approach reflects research showing that adult learners often learn most effectively through active participation, reflection, and authentic application.

Recommendations for Future Workshops

Based on survey findings and participant feedback, several recommendations emerged for improving future workshop offerings:

- Provide optional pre-workshop suggested readings for foundational math, statistics, and programming concepts.
- Include additional reflection and discussion periods to help learners consolidate difficult concepts.
- Divide the block lecture structure into smaller times
- Provide additional examples and scaffolded exercises for advanced concepts such as dimensionality reduction and matrix-based thinking.

These recommendations will help strengthen accessibility, learner confidence, and instructional effectiveness in future workshops.

Conclusion

Overall, the *AI for Advanced Semiconductor Manufacturing Workshop* demonstrated strong effectiveness in supporting adult learner engagement, confidence development, and applied technical learning. The integration of hands-on activities, industry-relevant applications, and guided instruction created a learning environment that aligned closely with principles of adult education and experiential learning.

Survey findings suggest that participants valued opportunities to work directly with real-world semiconductor manufacturing and AI applications while receiving support from knowledgeable and approachable instructors. The workshop successfully fostered learner confidence, practical skill development, and interest in continued technical education and workforce pathways.

At the same time, the evaluation identified several opportunities for improvement, particularly in supporting learners with limited technical backgrounds through additional scaffolding, preparatory resources, and structured reflection opportunities. These findings will inform future revisions of the workshop design and contribute to the continued development of inclusive and effective technical education practices.

From a teaching and learning perspective, this evaluation reinforced the importance of:

- balancing guidance with independent exploration,
- integrating experiential learning opportunities,
- and designing instruction that remains responsive to diverse adult learner needs.